NED University of Engineering and Technology, Karachi.
Department of Software Engineering (CS)
Midterm Examination 2024
Academic Reading & Writing EA-115
PAPER A

TIME: 90 MINS MARKS: 20

CT-23025

Q1. Read the following passage and answer the questions below. [CLO 1] Marks 10

Reading Passage

The Evolution of Software Engineering

A The field of software engineering has seen rapid development since its inception in the late 1960s. With the increasing complexity of software systems and the growing need for efficient, error-free applications, software engineers have developed methodologies and practices to improve both the quality and speed of software development. One such approach is agile software development, which emphasizes iterative progress, customer collaboration, and flexibility.

B Traditional software development models, such as the Waterfall model, operate in a linear fashion, where each phase must be completed before moving on to the next. While this approach ensures thorough documentation and planning, it has been criticized for its rigidity and inability to adapt to changing client needs. Agile, on the other hand, operates in short cycles, allowing for regular feedback and adjustments throughout the development process.

C Another crucial trend in modern software engineering is the adoption of DevOps practices. DevOps integrates software development (Dev) with IT operations (Ops) to improve collaboration, speed, and reliability in the software release process. Continuous integration, continuous delivery (CI/CD), and automation are key aspects of DevOps, making it a valuable practice in today's fast-paced development environments.

D Despite the benefits, these practices are not without their challenges. Agile projects can sometimes lack clear direction if customer feedback is inconsistent, and DevOps can be difficult to implement without proper tools and team alignment. However, with the right training and resources, software engineers can effectively balance flexibility and structure to deliver high-quality software solutions.

Question 1: Matching Information

(For each of the following statements, **choose the correct section (A, B, or C or D)** from the passage where the information is mentioned.)

1. Agile development allows for frequent client feedback. (1 mark)

2. DevOps combines development and operations for faster software release. (1 mark)

3. Traditional development models follow a strict sequence of steps. (1 mark)

Question 2: True/False/Not Given

(Do the following statements agree with the information in the passage? Write True if the statement agrees with the passage, False if it contradicts the passage, and Not Given if there is no information on this.)

4. The Waterfall model is highly adaptable to client needs. (1 mark)

5. Continuous integration is a key part of DevOps practices. (1 mark)

Question 3: Sentence Completion

(Complete the following sentences using NO MORE THAN TWO WORDS from the passage.)

6. Traditional software models are often criticized for their _____. (1 mark)

7. Agile software development follows a process of _____ cycles. (1 mark)

Question 4: Multiple Choice

(Choose the correct option based on the passage.)

8. One key difference between Agile and Waterfall models is: (1 mark)

a) Agile focuses more on documentation than Waterfall

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b) Waterfall allows for frequent feedback

c) Agile is more flexible and allows changes during development

d) Waterfall is faster than Agile

Question 5: Short Answer

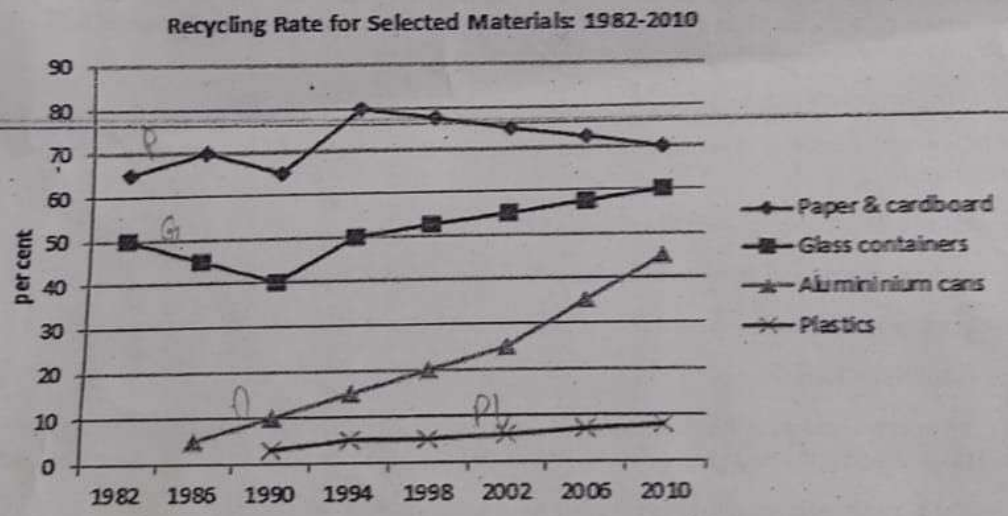
(Answer the following question using NO MORE THAN THREE WORDS from the passage.)

9. What is one key practice of DevOps that helps speed up software release? (1 mark)

10. What can cause problems in Agile projects? (1 mark)

Q2. Take a look at the chart below and perform the task as outlined: [CLO 2] Marks : 06

This graph shows the proportion of four different materials that were recycled from 1982 to 2010 in a particular country. **Summarise the information** by selecting and reporting the main features, making comparisons where relevant. **100- 150 words**



Q3. Question: In-Text Citation (APA Style) **[CLO 3] Marks: 04**

You are writing a research paper and want to include the following information from a journal article in APA 7th edition style:

- Author: Davis, M. J.
- Year: 2021
- Title of Article: "The Role of Automation in Modern Software Engineering"
- Journal Name: *Journal of Software Innovation*
- Volume: 12
- Issue: 3
- Pages: 45-60
- DOI: 10.1234/jsi.2021.1203

Task: Text from the article:

"Automation in software development has revolutionized the industry by enhancing efficiency and reducing human error."

1. Paraphrase the above sentence and include a correct in-text citation in APA format. (2 marks)
2. Use the same sentence as a direct quote with a proper in-text citation in APA format. (2 marks)